

Salt Stress

ReLEAF Standardized Salinity Stress & Protectant Application Protocol

Purpose

To evaluate whether candidate plant protectants improve *Arabidopsis thaliana* performance under NaCl-induced salinity stress using standardized growth, treatment, imaging, and measurement conditions.

1. Experimental Design

Use seedlings of similar developmental stage and root length.

Standardized Agar Set up/formula

- Arabidopsis thaliana Columbia (Col-0 accession)
- Circular horizontal plates / Vertical plates planting
- Half-strength Murashige and Skoog medium (½ MS medium)
- 1% sucrose

Standardized experimental preparation set ups:

- Vernalize 4 days in darkness at 4°C
- Incubate in growth chamber for 4 days germination under standardized growth condition

2. Prepare ½ MS Growth Medium

For 1 L of agar solution:

Component	Amount
½ MS medium powder	2.3 g
MES buffer	0.5 g
Sucrose	10 g
Agar	10 g

Distilled water	To 1 L
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- Mix components thoroughly using a hot plate to stir.
- Adjust to **pH 5.7–5.8** using 0.1 M KOH if necessary.
- Autoclave for 20 minutes at 127°C
- Pour corresponding:
 - 30ml agar for circular horizontal plates
 - 50ml agar for vertical plates

3. Prepare *Arabidopsis*

For Vertical plates

- Use the seed template (3cm, 5cm, 4cm) to plant 18 seeds on each agar plate in two rows, nine seeds on each row
- Use breathable tape to seal each of the agar plates
- Label the vernalization date
- Wrap the agar plates with aluminum foil
- Vernalize seeds at **4°C in darkness for 4 days**.
- Incubate in the growth chamber for 4 days germination under standardized growth conditions
- Select seedlings with similar growth and root length.
- Draw a horizontal reference line approximately **3 cm from the top of the square plate**.
- Transfer seedlings based on the template so the root starting points align with the line. (The position allows subsequent root growth to be measured consistently)
- Grow plates vertically with a slight backward tilt around 15°.

For Circular horizontal plates

- Use the circular horizontal plate seed template to plant seeds on each agar plates (number of seeds vary upon the trial settings)
- Use breathable tape to seal each of the agar plates
- Label the vernalization date
- Wrap the agar plates with aluminum foil
- Vernalize seeds at **4°C in darkness for 4 days**.
- Incubate in the growth chamber for 4 days germination under standardized growth conditions

4. Application Scenarios (Salinity stress & Protectants)

Three Application Methods Based on Protectants

1. Trehalose

Application: Mix with Agar mixture before solidification:

- **1/2 MS agar plate preparation(salinity test trial plates)**

The following protocol is for 600ml of agar

- 12 agar plates
 - control(no trehalose) x control(no NaCl)
 - control(no trehalose) x 150mM NaCl
 - control(no trehalose) x 250mM NaCl
 - 1mM trehalose x control(no NaCl)
 - 1mM trehalose x 150mM NaCl
 - 1mM trehalose x 250mM NaCl
 - 5mM trehalose x control(no NaCl)
 - 5mM trehalose x 150mM NaCl
 - 5mM trehalose x 250mM NaCl
 - 10mM trehalose x control(no NaCl)
 - 10mM trehalose x 150mM NaCl
 - 10mM trehalose x 250mM NaCl
- Prepare the following:

½ MS medium powder(w/o sucrose)	1.38g
MES buffer	0.3g
Distilled water	600ml
Sucrose(1%)	6g
Agar powder(1%)	6g

- Add the contents into a 1000ml flask
- Use a hot plate to stir (with a magnetic stirring rod) until no sediments are left in the solution
- Use the pH probe and measure the pH of the solution, make sure the pH is 5.7-5.8
 - If the pH is too low, add 0.1M KOH slowly to increase it
- Pour 200ml of mixture out into another flask to separate into three flasks, label with [0mM], [150mM] and [250mM]

- Add 2.192 g of NaCl into the flask labelled [150mM NaCl]
- Add 3.652 g of NaCl into the flask labelled [250mM NaCl]
- Prepare 4 flasks for different trehalose concentrations(0, 1, 5, 10mM)
- Prepare 12 flasks and pour the respective trehalose and NaCl concentration in each of the flasks
 - making sure there is 50ml in each of the flasks(12)
 - label each flask clearly according to their trehalose concentrations
- To their respective trehalose concentrations, add

Trehalose concentration	Amount of trehalose (for 50ml of medium)
0mM	0μL
1mM	500μL
5mM	2500μL
10mM	5000μL

- Wrap the necks of the flasks with aluminum foil and stick a piece of indicator tape on top
- Autoclave for 20 minutes at 127°C
 - Wait for the pressure to reach 0 before retrieving the flask
- During the autoclave, label 8 of the petri dishes with a marker in the biosafety cabinet
 - right top: date and arrow pointing up to show direction for pictures
 - bottom lid: different trehalose concentrations and NaCl concentration or control
- Transfer half of the flasks to BSC
 - Store the other half of the flasks in the hot bath while pouring so that it does not solidify
- Make sure its surface is dry by wiping the flasks with a paper towel before spraying 70% EtOH on them
- Pour each of the medium mixtures into their square petri dishes (around 50ml)
- Transfer the other half of the flasks into the BSC
- Pour each of the medium mixtures into their square petri dishes
- Let the petri dishes solidify at room temperature under the BSC

2. CH bio peptide

Application: Top Drench solution on Agar Surface with Treatment Timings

The same protocol can be used to investigate **when** protectants should be released.

Treatment	Timing
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Pretreatment	Protectant applied 1 day before NaCl stress
Cotreatment	Protectant applied at the same time as NaCl stress
Post-treatment	Protectant applied 1 day after NaCl stress

(After plants have been grown on agar in growth chamber for 4 days)

- Dilute NaCl solutions and CH bio peptide
 - Dilution factor of NaCl solutions based on concentrations

NaCl concentration	Dilution factor
0mM	0µL
75mM NaCl	500µL
100mM NaCl	2500µL
150mM NaCl	5000µL

- Dilute CH bio peptide by 1/500 dilution rate into a 15ml centrifuge tube

→ 9.98ml of sterile ddH₂O + 20ul of peptide

- Treatment Timings:
 - Apply protectants to pre-treatment groups (1 day before)
 - Aspirate 1ml of diluted peptide and apply onto the agar surface (ensure solution has submerged the plants)
 - Apply protectants and NaCl to co-treatment groups (1 day before)
 - Aspirate 1ml of diluted peptide and apply onto the agar surface (ensure solution has submerged the plants)
 - Aspirate 1ml of specific NaCl concentration into its corresponding group (all treatment groups receive salinity stress at the same day)
 - Apply protectants and NaCl to after-treatment groups (1 day after)
 - Aspirate 1ml of diluted peptide and apply onto the agar surface (ensure solution has submerged the plants)

3. ACCD

Application: foliar spraying

(After plants have been grown on agar in growth chamber for 4 days)

- Purify ACCD and perform buffer exchange/dialysis before plant application.

- Transfer the ACCD solution into a clean spray container.
- Treatment Timings:
 - Spray CH bio peptide on pre-treatment plate (1 day before)
 - Open the BSC
 - Dilute the CH bio peptide by 1/500 dilution rate (19.96ml of sterile ddH₂O + 40ul of peptide) into a 50ml centrifuge tube
 - Apply an individual sprayer onto the centrifuge tube
 - Take out the plate labeled “pre-treatment”, place it vertically, spray the CH bio peptide toward the plants entirely from leaves to roots with a 15cm distance in between (around 2 sprays, make sure all the plants received treatment)
 - Seal the plates with breathable tape and put it back into the growth chamber
 - Spray CH bio peptide on co-treatment plate (day 0)
 - Open BSC
 - Take out the plate labeled “co-treatment”, place it vertically, spray CH bio peptide toward the plants entirely from leaves to roots with a 15cm distance in between (around 2 sprays, make sure all the plants received treatment)
 - Seal the plates with breathable tape
 - Place all 4 groups of plates onto a rack while the plates stand with a slightly tilted angle, then place the rack entirely into the incubator at 37°C under white light for thermo priming
 - Observe for 5 days inside the incubator
 - Spray CH bio peptide on after-treatment plate (1 day after)
 - Open BSC
 - Take out the plate labeled “after-treatment” from the incubator, place it vertically, spray CH bio peptide toward the plants entirely from leaves to roots with a 15cm distance in between (around 2 sprays, make sure all the plants received treatment)
 - Seal the plates with breathable tape and place it back into the incubator

Note that the **0 mM NaCl group** receives the same volume of sterile water treatment as the protectant/NaCl to ensure the equivalent amount of liquid exposed to the plants across groups.

ACCD concentration: _____

Volume per spray: 1.6667ml

LEA14

Application method: purified protein solution

For consistency with the ReLEAF salinity-testing platform, apply purified LEA14 externally to the plant rather than incorporating engineered bacteria into the plant experiment.

- Prepare purified LEA14 in the selected low-salt plant-compatible buffer.
- Apply the same volume to each treatment group.
- Apply buffer alone to corresponding control groups.

Recommended standardized delivery format for comparison: foliar application using the same spray volume and number of sprays as the ACCD trial.

LEA14 concentration: _____

Number of sprays: _____

Volume per spray: _____

Exact LEA14 dose has not yet been established in the supplied experimental records and should be experimentally optimized before being presented as a validated concentration.

BoPep4

Application method: purified peptide solution

- Prepare BoPep4 at the experimentally selected working concentration.
- Apply an equal volume to plants in each BoPep4 treatment group.
- Use the corresponding peptide buffer without BoPep4 as the treatment control.

To keep comparisons between ReLEAF protectants consistent, use the **same foliar spray framework** used for ACCD unless a different delivery method is experimentally shown to perform better.

BoPep4 concentration: _____

Number of sprays: _____

Volume per spray: _____

Exact BoPep4 application concentration is not defined in the supplied plant-testing records and should remain listed as an optimization parameter until experimentally validated.

6. Treatment Timing

The same protocol can be used to investigate **when** protectants should be released.

Treatment	Timing
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Pretreatment	Protectant applied 1 day before NaCl stress
Cotreatment	Protectant applied at the same time as NaCl stress
Post-treatment	Protectant applied 1 day after NaCl stress

This structure was used in later protectant experiments to compare timing-dependent rescue effects.

For a simple first comparison, use **cotreatment** for all three protectants.

7. Monitor Plant Response

Observe plants for approximately **10 days**.

Take standardized photographs daily using:

- the same black background
- the same lighting
- the same plate orientation
- the same camera distance + quality control

Record:

Daily

- root growth
- leaf area / plant size
- visible phenotype
- survival

Data collection

- freshweight
- chlorophyll content
- final root length

For protectant testing groups

- MDA
- Total protein
- Enzymatic activity testing

Experimental Logic

****Salt only**

**** → determines the effect of salinity stress**

****Protectant only**

**** → determines whether the protectant itself affects plant growth**

****Salt + protectant**

**** → determines whether the protectant rescues plants from salinity stress**

****Pretreatment vs. cotreatment vs. post-treatment**

**** → determines when protectant delivery is most effective**

Heat Stress

ReLEAF Standardized Heat Stress & Protectant Application Protocol

Purpose

To evaluate whether candidate plant protectants improve *Arabidopsis thaliana* tolerance or recovery under controlled heat stress using standardized growth, treatment, imaging, and measurement conditions.

1. Experimental Design

Use seedlings of similar developmental stage and root length.

Group	Heat Stress	Protectant
Negative Control	No	None
Protectant Control	No	Protectant
Heat Control	Yes	None
Heat + Protectant	Yes	Protectant

2. Prepare ½ MS Growth Medium

For 1 L:

Component	Amount
½ MS medium powder	2.3 g
MES buffer	0.5 g
Sucrose	10 g
Agar	8–10 g
Distilled water	To 1 L

- Mix all components thoroughly.
- Adjust to **pH 5.7–5.8** using 0.1 M KOH if needed.
- Sterilize using the laboratory’s standard agar-medium procedure.
- Pour approximately **30–50 mL** into each square plate.
- Allow plates to solidify completely.

3. Prepare *Arabidopsis*

- Vernalize seeds at **4°C in darkness for 4 days**.
- Germinate seedlings under standardized growth conditions.
- Select seedlings with similar:
 - developmental stage
 - germination status
 - root length
- Draw a horizontal reference line approximately **3 cm from the top of the square plate**.
- Transfer seedlings so that the starting points of the roots align with the line.
- Grow plates vertically with a slight backward tilt.

This keeps initial plant position consistent between treatment groups.

4. Apply Heat Stress

Use one standardized heat-stress condition for all treatment groups within the same experiment.

Two formats can be tested:

Long-Term Heat Stress

Plants are exposed to an elevated temperature for an extended period.

Temperature: _____ °C

Exposure duration: _____ hours / days

Short-Term Heat Shock

Plants are exposed to a higher temperature for a shorter period before being returned to normal growth conditions.

Temperature: _____ °C

Exposure duration: _____ minutes / hours

Keep the following identical between control and treatment groups:

- plant age
- plate type
- growth medium
- light conditions
- humidity
- duration of exposure
- recovery time

The exact temperature and duration should be entered based on your team's validated heat-stress experiment rather than presenting an untested value as standardized.

5. Protectant Application

ACC Deaminase — ACCD

Application method: foliar spray

- Purify ACCD and perform buffer exchange/dialysis before plant application.
- Prepare ACCD in a plant-compatible low-salt buffer.
- Transfer the solution into a clean spray container.
- Apply an equal number of sprays to each treatment group.
- Apply the corresponding buffer without ACCD to control plants.

ACCD concentration: _____

Number of sprays: _____

Volume per spray: _____

For consistency with the salinity protocol, the same foliar delivery approach can be used when comparing heat-stress responses.

LEA14

Application method: purified protein solution

Because LEA14 is being evaluated as a stress-protective protein, apply it externally to plants in a standardized manner.

- Prepare purified LEA14 in the selected plant-compatible buffer.
- Apply the same concentration and volume to every treatment plant.
- Apply buffer alone to the matched control group.

For cross-protectant comparison, use the same foliar spray format as ACCD unless another delivery method has been experimentally validated.

LEA14 concentration: _____

Number of sprays: _____

Volume per spray: _____

LEA14 is especially appropriate to prioritize in this assay because the experimental question is whether it improves plant performance during or after heat stress.

BoPep4

Application method: purified peptide solution

- Prepare BoPep4 at the selected working concentration.
- Apply an equal volume to every treatment plant.
- Treat control plants with the same carrier buffer without BoPep4.

Use the same foliar spray framework where possible so results can be compared across protectants.

BoPep4 concentration: _____

Number of sprays: _____

Volume per spray: _____

If BoPep4 is primarily being tested for another stress type, this heat-stress assay should be described as exploratory rather than its primary validation assay.

6. Treatment Timing

The same framework can determine when a protectant is most effective.

Treatment	Timing
Pretreatment	Protectant applied 1 day before heat stress
Cotreatment	Protectant applied immediately before or at the start of heat stress
Post-treatment	Protectant applied after heat exposure , during recovery

For an initial comparison, use **cotreatment** across all protectants.

For a more complete experiment, compare all three timings.

7. Recovery

After heat treatment:

- Return plants to the standard growth condition.
- Keep recovery conditions identical for all groups.
- Record the recovery period.

Recovery temperature: _____ °C

Recovery duration: _____ days

Recovery measurements are especially useful because some protectants may not completely prevent heat damage but may improve the plant's ability to recover afterward.

8. Monitor Plant Response

Observe plants for approximately **7–10 days**, or according to the validated experimental design.

Daily

Take standardized photographs using:

- same black background
- same lighting
- same camera position
- same plate orientation

Record:

- leaf appearance
- wilting
- plant size
- root growth
- leaf area
- survival
- visible recovery

Endpoint measurements

Measure:

- fresh weight
- chlorophyll content
- root length
- leaf area
- survival percentage
- visible heat-damage phenotype

Experimental Logic

**No heat + no protectant

**→ normal growth baseline

**No heat + protectant

**→ determines whether the protectant affects plants without stress

**Heat + no protectant

**→ measures the effect of heat stress alone

**Heat + protectant

**→ tests whether the protectant improves heat tolerance or recovery

**Pre vs. Co vs. Post-treatment

**→ identifies the most effective time for protectant delivery

Mix with agar mixture before solidification

Divide the 1000 ml mixture equally into the number of flasks regarding how many distinct NaCl concentrations are included

Eg.

For an experiment of 0, 150, 250mM NaCl vs. 0, 1, 5, 10mM Trehalose:

- Split into NaCl Groups:
- Divide the 1000 ml mixture equally into 3 flasks ($1000/3=333\text{ml}$ (approximate) per flask).
- Label the flasks: [0 mM NaCl], [150 mM NaCl], and [250 mM NaCl].
- Add NaCl:
- [0 mM NaCl] flask: Add 0 g of NaCl.
- [150 mM NaCl] flask: Add 2.92 g of NaCl.
- [250 mM NaCl] flask: Add 4.87 g of NaCl.
- Stir each flask until the salt is completely dissolved.